USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 7 LAB - AM and FM Receivers
Lab Packet
Spring 2027

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 3.3: I can implement an FM receiver in GNU Radio, demodulate a live broadcast FM signal, and decode its RDS data.

Learning Objective 3.4: I can explain and demonstrate the trade-off between bandwidth and noise performance for AM vs. FM.

This packet is the L7 Lab, in three parts, and you turn it in once, complete, before Lesson 8. Do **Part 1** on your own before the lab period: predict the receiver settings and the numbers you will measure. **Part 2** is your own measurements, on your own laptop and RTL-SDR, recorded in class. Do **Part 3** on your own after the lab: compare the measurements with the predictions and answer the analysis questions.

Grading. We grade one submission against Objectives 3.3 and 3.4 on the course's 1 / 0 scale, and you may revise it under course policy. Paste screenshots where a question asks for one.

Receive only. Every flowgraph in this lab receives or simulates. The RTL-SDR cannot transmit, and nothing in this lab needs a license. Start every receiver with the volume low.

Files. Download the six L7 flowgraphs from the L7 lesson page into one folder, with your `fm_capture.cfile` from the L5 Lab. Open `L07-rds-rx.grc` first and confirm that GRC shows no missing blocks: gr-rds comes with radioconda; if GRC reports the RDS blocks missing, run `mamba install gnuradio-rds` from the radioconda prompt. Then, before the period, run `L07-fm-test-tx.grc`, which also uses an RDS block, and `L07-am-test-tx.grc` once each; they write `L07-fm-test.cfile` and `L07-am-test.cfile` into that folder and stop on their own.

Part 1: Pre-Lab (individual, before the lab period)

1. **Predictions.** You will check each of these at the bench.

   (a) Fill in the receiver settings for the test file and for your own L5 capture. The shift is the Frequency Xlating FIR Filter's center frequency.

Recording	SDR center (MHz)	Station (MHz)	Shift (kHz)
`L07-fm-test.cfile`	100.0	100.3	
`fm_capture.cfile`			

   (b) The FM receiver decimates 2.4 MS/s by 10 and then by 5. Find the two rates, and the Quadrature Demod gain that makes 75 kHz of deviation an output of 1.

gain = _____

(c) The test station's audio is a 1 kHz tone at amplitude 0.6 plus a 2.5 kHz tone at 0.4, with pre-emphasis at the transmitter and de-emphasis at the receiver. Predict how far below the 1 kHz line the 2.5 kHz line sits on the receiver's audio sink.

(d) RDS sends 1187.5 bits per second in 104-bit groups, and the station name takes four groups. Find the shortest time from a clean start to a displayed name.

(e) Find the ATIS frequency of an airport near you in the FAA Chart Supplement or on a sectional chart. Choose an SDR center 250 kHz below it, and fill in the AM receiver's settings.

Airport	ATIS (MHz)	SDR center (MHz)	Shift (kHz)	Decimation

(f) In `L07-am-vs-fm.grc` the FM channel is $2(25 + 5) = 60$ kHz wide and the AM channel 10 kHz. FM reaches threshold at about 10 dB C/N in its own bandwidth. Predict the C/N in 10 kHz at which the 25 kHz FM signal reaches threshold.

Part 2: Measurements (your own laptop and RTL-SDR, in class)

2. **Build the WBFM receiver.** Start GRC with an empty canvas. Do not open `L07-wbfm-rx.grc`: build the receiver yourself.

   - `samp_rate` 2.4e6, a QT GUI Entry `fc_mhz` (100.0), a QT GUI Entry `station_mhz` (100.3), and a QT GUI Range `vol` from 0 to 1.
   - A File Source on `L07-fm-test.cfile`, repeat on.
   - A Frequency Xlating FIR Filter, decimation 10, center frequency `(station_mhz - fc_mhz) * 1e6`, with a Low-pass Filter Taps variable `chan_taps`: cutoff 105 kHz, transition 30 kHz, at `samp_rate`.
   - A Quadrature Demod with the gain of Question 1(b).
   - A Decimating FIR Filter, decimation 5, with taps `firdes.low_pass(1, samp_rate/10, 15e3, 3e3)`.
   - FM Deemphasis at 48 kHz with $\tau = 75$ μs, a Multiply Const `vol`, an Audio Sink at 48 kHz, and a float QT GUI Frequency Sink on the de-emphasis output. Do not add a Throttle.

   (a) Paste a screenshot of your flowgraph on the GRC canvas. Every connection must be black, not red.

   (b) Run it. Read the audio frequency sink, and record the frequency of each line and how far the 2.5 kHz line sits below the 1 kHz line.

   (c) Type 99.5 into Station. Describe what you hear, and explain why the station at 100.3 MHz disappears.

3. **Your L5 capture.** In your receiver from Question 2, set the File Source to `fm_capture.cfile`, SDR center to the frequency you recorded it at, and Station to the station you centered. Add a float QT GUI Frequency Sink on the Quadrature Demod output, with bandwidth `samp_rate/10`.

   (a) Paste a screenshot of the demodulated multiplex. Mark each part of the broadcast multiplex you can see: mono audio, the 19 kHz pilot, the stereo band from 23 to 53 kHz, and RDS at 57 kHz.

(b) Describe the audio. Your capture is 2 s long, and the File Source repeats it.

4. **Live FM and RDS.** Open `L07-rds-rx.grc`. Disable the File Source, enable the RTL-SDR Source, and set SDR center about 300 kHz below a strong local station and Station to the station. Set RF gain from your L5 Lab gain table.

   (a) Record what the RDS panel shows. The panel prints the PI code under "Program Type" and the program type under "PI"; record each under its correct name.

Station (MHz)	
Station name (PS)	
PI code (hex)	
Program type	
Radiotext	

   (b) Paste a screenshot of the RDS constellation and the panel. Explain what the two clusters show about the receiver.

   (c) Restart the flowgraph and time how long the station name takes to appear. Compare with Question 1(d).

5. **ATIS.** Open `L07-am-rx.grc`. Check it first on `L07-am-test.cfile`, then disable the File Source, enable the RTL-SDR Source, and enter the settings of Question 1(e).

   (a) Paste a screenshot of the receiver on the live ATIS, and record the information letter and the altimeter setting it broadcasts.

   (b) Lower RF gain until the voice is hard to understand, and describe how the AM audio degrades. You

will compare it with FM in Question 6(b).

6. AM and FM in the same noise. Open and run `L07-am-vs-fm.grc`. It needs no hardware.

(a) Read the two output SNR meters at each C/N, with the FM deviation at 25 kHz and then at 5 kHz. Near threshold the meters wander; record the middle of the range you see.

C/N in 10 kHz (dB)	30	25	20	15	10
AM					
FM, 25 kHz					
FM, 5 kHz					

(b) Set the deviation to 25 kHz and Listen To to FM. Lower C/N from 30 to 10 dB and describe the sound at 20, 15, and 10 dB.

Part 3: Writeup (individual, after the lab)

7. Analysis.

(a) Your mono receiver plays a stereo station correctly. Using your multiplex screenshot from Question 3(a), explain why.

(b) Compare where your 25 kHz FM measurements lost their advantage over AM with your prediction in Question 1(f). Then use your table to explain why the 5 kHz signal has the higher output SNR at 10 dB C/N and the lower one at 30 dB.

(c) Aircraft use AM, and broadcasters use wideband FM. Using your measurements from Questions 5 and 6, explain one reason for each choice.